Interactive Cabinet

Technical Usage Documentation

Blueprint package for Unreal Engine featuring interactive cabinets (doors and drawers) ready to use in laboratory scenes, common household interiors, or general architectural visualization.

Document version: 1.0

Engine: Unreal Engine 5.6

Package contents: BP_Cabinet1, BP_Cabinet2

1. Overview

This package contains two cabinet Blueprints — **BP_Cabinet1** and **BP_Cabinet2** — designed for laboratory scenes or common household interiors. Both use a **PoseableMesh** component as their base, which allows each door and drawer to be animated individually without requiring complex Skeletal Meshes or pre-baked animations.

The opening of each door or drawer is controlled by **Float** variables, ranging from **0.0** (fully closed) to **1.0** (fully open). These values can be adjusted directly in the editor (Details Panel of the actor placed in the level) or at runtime through the **OpenDrawer** and **OpenDoor** functions.

1.1 Differences between the two Blueprints

Blueprint	Drawers	Doors	Main functions	Notes
BP_Cabinet1	Up to 10 (Drawer1–Drawer10)	Multiple (Door1, Door2...)	Construction Script, OpenDrawer, OpenDoor	Includes Event BeginPlay, Event ActorBeginOverlap and Event Tick, ready for customization.
BP_Cabinet2	—	Up to 4 (Door1–Door4)	Construction Script, OpenDoor	Door-only variant; includes Material and MaxDistance (Vector) variables for visual/physical adjustment.

2. Requirements

- Unreal Engine 5 (pure Blueprint, no third-party plugin dependencies).
- No additional software is required to use this package — all content is native Blueprint.
- Basic familiarity with the Unreal Engine editor (Content Browser, Details Panel and Level Editor) is recommended.

If you have no prior experience with Unreal Engine, Epic Games provides free official documentation at https://dev.epicgames.com/documentation/unreal-engine, including guides on Blueprints Visual Scripting and the Level Editor.

3. Blueprint Internal Structure

Both Blueprints follow the same component logic and variable organization, making it easier to maintain and to create new cabinets based on the same pattern.

3.1 Components

- **DefaultSceneRoot** — the actor's root, used as the reference point for transforms.

- **PoseableMesh** — the cabinet's main mesh; the bones corresponding to each door/drawer are moved individually through this component.

3.2 Functions

Function	Description
Construction Script	Runs when the actor is built/edited. Applies materials and prepares initial references for the PoseableMesh.
OpenDrawer	Takes a Float value (0.0 to 1.0) and the drawer index; moves the corresponding PoseableMesh bone proportionally to the given value. Available only in BP_Cabinet1.
OpenDoor	Takes a Float value (0.0 to 1.0) and the door index; rotates the corresponding PoseableMesh bone proportionally to the given value. Available in both BP_Cabinet1 and BP_Cabinet2.

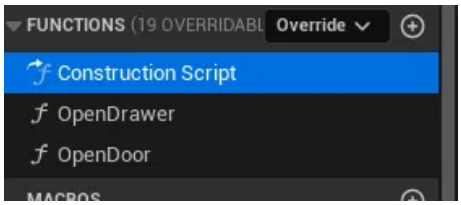


Figure 1 — Functions panel in My Blueprint, showing Construction Script, OpenDrawer and OpenDoor.

3.3 Control variables

Each door and drawer has its own independent Float variable, grouped under the **Drawer Open** and **Door Open** categories. Each variable's value represents the opening percentage, from 0.0 (closed) to 1.0 (fully open).

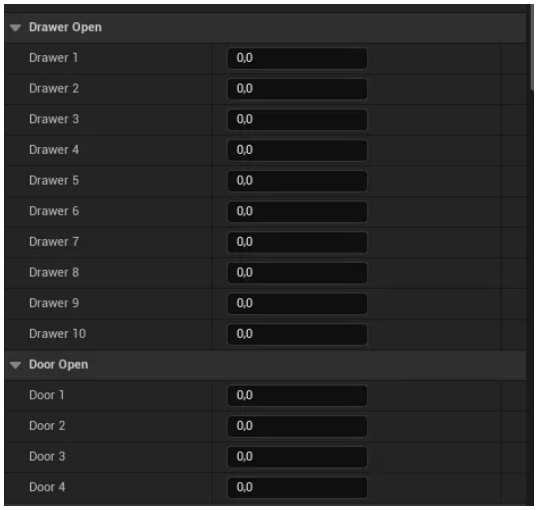


Figure 2 — Drawer1–Drawer10 and Door1–Door4 variables in the Details Panel, with values at 0.0 (closed).

Category	Variables	Type	Range
Drawer Open	Drawer1 ... Drawer10	Float	0.0 to 1.0
Door Open	Door1 ... Door4 (or as configured)	Float	0.0 to 1.0

In BP_Cabinet2, in addition to the door variables, there are two supporting variables: **Material** (Material Interface, used to swap the cabinet's finish) and **MaxDistance** (Vector, used as a reference for the PoseableMesh's maximum movement limit).

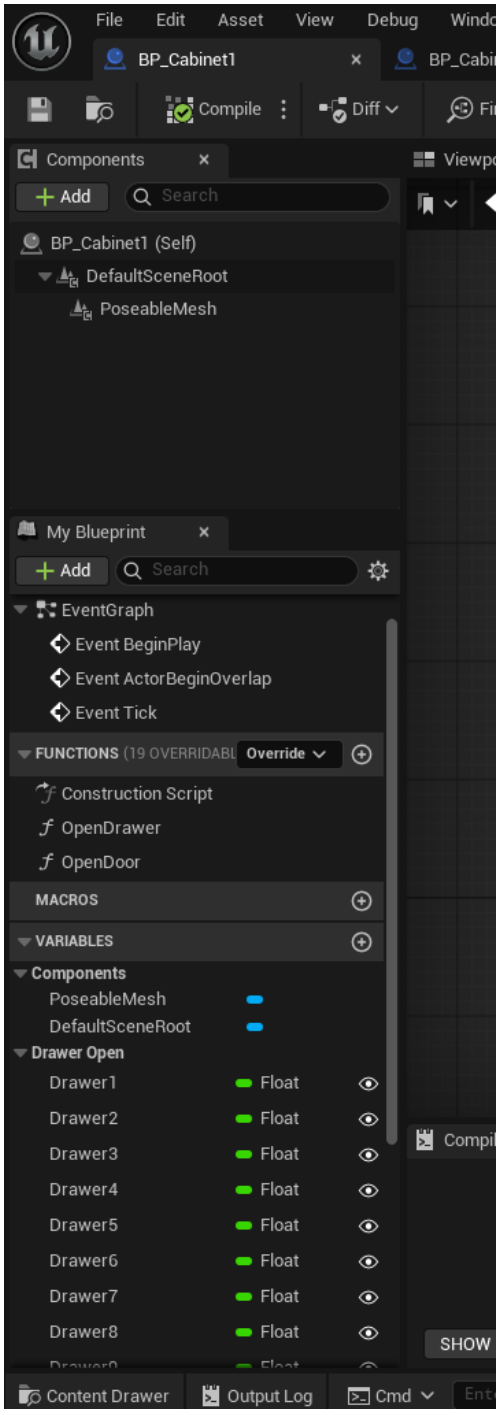


Figure 3 — Full structure of BP_Cabinet1: components, events, functions and Drawer Open variables.

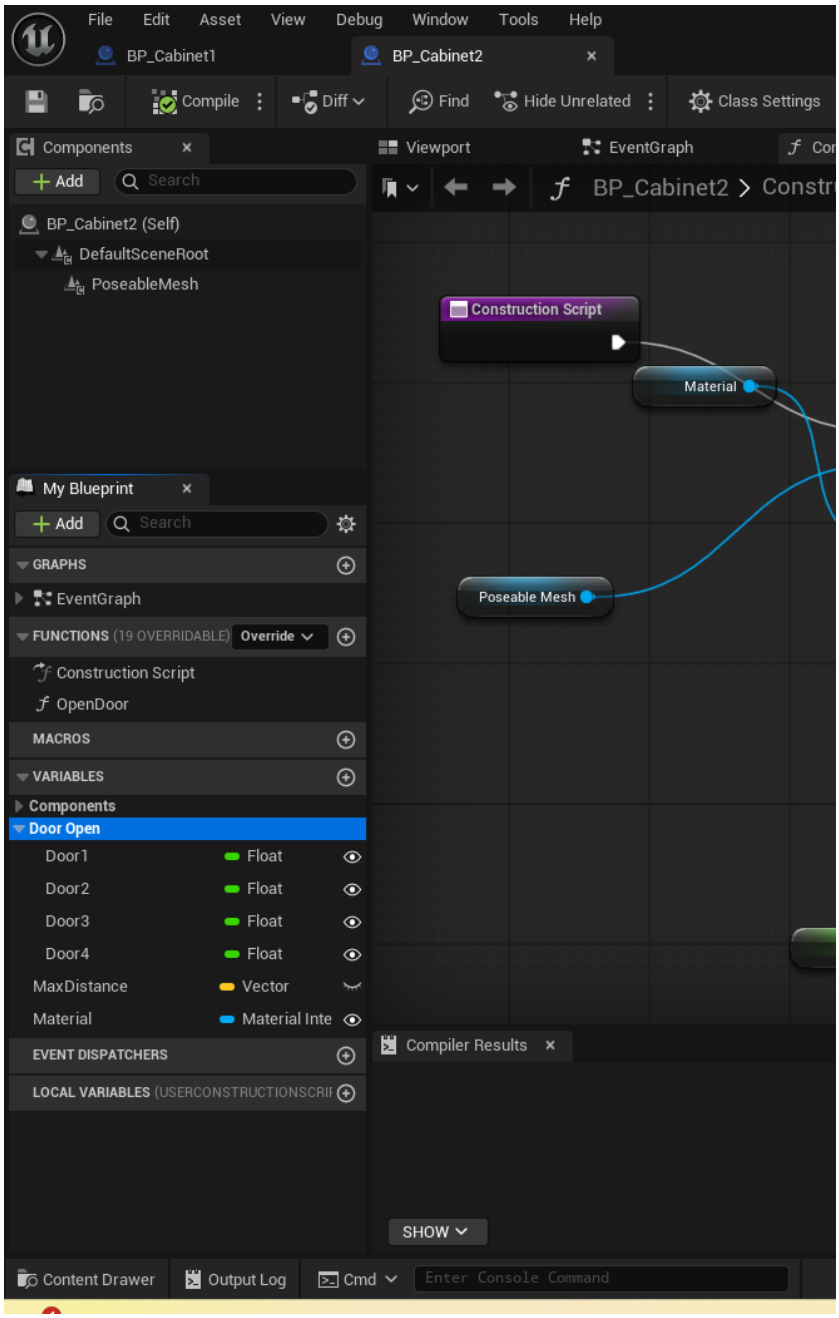


Figure 4 — Structure of BP_Cabinet2, with the Construction Script connecting Material and Poseable Mesh, and the Door Open variables.

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4. Materials & Textures

The package includes a single **Master Material** driving four **Material Instances**, one for each available finish. Switching between finishes only requires assigning a different Material Instance to the cabinet — no additional setup is needed.

4.1 Available finishes

- Cedar
- Cherry Wood

- Galvanized Steel
- Oak

4.2 Texture maps

Each finish provides a full PBR texture set, following the **BC** (Base Color), **N** (Normal) and **ORM** (Occlusion / Roughness / Metallic) naming convention:

Finish	Base Color (BC)	Normal (N)	ORM
Cedar	T_Cabinet_Cedar_BC	T_Cabinet_Cedar_N	T_Cabinet_Cedar_ORM
Cherry Wood	T_Cabinet_CherryWood_BC	T_Cabinet_CherryWood_N	T_Cabinet_CherryWood_ORM
Galvanized Steel	T_Cabinet_GalvanizedSteel_BC	T_Cabinet_GalvanizedSteel_N	T_Cabinet_GalvanizedSteel_ORM
Oak	T_Cabinet_Oak_BC	T_Cabinet_Oak_N	T_Cabinet_Oak_ORM

4.3 Applying a finish

- Select the cabinet actor in the level.
- In the Details Panel, assign the desired Material Instance (Cedar, Cherry Wood, Galvanized Steel or Oak) to the PoseableMesh's material slot.
- In BP_Cabinet2, the finish can alternatively be set through the Material variable exposed in the Construction Script, allowing the finish to be changed per-instance directly in the Details Panel without editing the Blueprint.

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5. How to Use

4.1 Adding the cabinet to a scene

- Locate BP_Cabinet1 or BP_Cabinet2 in the Content Browser.
- Drag the desired Blueprint into the level (Level Viewport).
- Position, rotate and scale the actor using the engine's standard tools.

4.2 Opening doors and drawers directly in the Editor

With the actor selected in the scene, open the **Details Panel** and locate the **Drawer Open** and **Door Open** categories. Simply type a value between 0.0 and 1.0 in the field for the desired door or drawer to preview the opening directly in the Viewport, without needing to run the game (Play).

4.3 Opening doors and drawers at runtime (via Blueprint)

To control the opening during gameplay, call the **OpenDrawer** or **OpenDoor** functions from another Blueprint (for example, a player interaction Blueprint), passing the element's index and the desired opening value (0.0 to 1.0).

Conceptual call example:

```
Get Reference to BP_Cabinet1 → OpenDrawer ( Index = 2 , Value = 1.0 )
Get Reference to BP_Cabinet2 → OpenDoor ( Index = 1 , Value = 0.5 )
```

- Value 0.0 = element fully closed.
- Value 1.0 = element fully open.
- Intermediate values (e.g. 0.5) allow partial opening, useful for smooth animations (Timeline) or gradual interactions.

4.4 Suggested smooth animation

For a smooth transition between open and closed, use a **Timeline** node in the Blueprint that calls OpenDrawer/OpenDoor, interpolating the value from 0.0 to 1.0 over time, instead of applying the final value instantly.

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6. Customization

- **Materials:** change the Material variable (BP_Cabinet2) or the material applied to the PoseableMesh to swap the cabinet's finish.
- **Number of doors/drawers:** new Float variables can be added following the same naming pattern (DoorN / DrawerN), as long as the corresponding bone exists in the PoseableMesh.
- **Movement limits:** the MaxDistance variable (BP_Cabinet2) can be adjusted to control the maximum allowed displacement.

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7. Support & Requirements

This package has no third-party software dependencies — it works entirely with native Unreal Engine Blueprint functionality.

For questions, troubleshooting, or feature requests related to this package, join our Discord support community:

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Discord: https://discord.gg/sb26qyvD
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Free official Unreal Engine documentation, for general questions about Blueprints and the editor:

- Unreal Engine Documentation: https://dev.epicgames.com/documentation/unreal-engine
- Blueprints Visual Scripting Guide: https://dev.epicgames.com/documentation/en-us/unreal-engine/blueprints-visual-scripting-in-unreal-engine

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